Insertion Sort:

Insertion Sort is a simple sorting algorithm that builds the sorted array one element at a time. It picks one element from the unsorted part and inserts it into its correct position in the sorted part.

Algorithm:

```
InsertionSort(arr)

1. Start
2. For i = 1 to n - 1
      a. key = arr[i]
      b. j = i - 1
      c. While j >= 0 and arr[j] > key
            arr[j + 1] = arr[j]
            j = j - 1
      d. arr[j + 1] = key
3. Stop
```

Output:

```
Enter the number of elements: 5
Enter the elements:
7
8
9
4
5

Sorted Array:
[4, 5, 7, 8, 9]

Execution Time: 0.0000221230 seconds

Time Complexity:
Best Case     : O(n)
Average Case : O(n²)
Worst Case    : O(n²)
Space Complexity: O(1)


...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity:

- **Best Case: O(n)**
  The array is already sorted, so only one comparison is needed for each element.
- **Average Case: O(n^2)**
  Each element is compared and shifted to its correct position.
- **Worst Case: O(n^2)**
  The array is in reverse order, so the maximum number of comparisons and shifts are performed.

Space Complexity:

- **Iterative Insertion Sort: O(1)**
  Uses only one extra variable (key) and a few index variables.